Chapter 7 — Reaction Kinetics

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Rate of reaction is change in: (a) Temperature/time (b) Volume/time (c) Mass/time (d) Concentration/time

2. Rate constant is independent of: (a) Temperature (b) Nature of reactants (c) Catalyst (d) Concentration

3. Effective collisions need: (a) Only energy (b) Low temperature (c) Energy + orientation (d) Only orientation

4. Activation energy is: (a) Bond energy (b) Ionization energy (c) Heat of reaction (d) Minimum energy

5. Higher temperature increases: (a) Activation energy (b) Molar mass (c) Fraction above Ea (d) Bond length

6. Catalyst increases rate by: (a) Changing ΔH (b) Lowering Ea (c) Removing products (d) Raising Ea

7. Powder reacts faster due to: (a) High density (b) Low temperature (c) More surface area (d) Less surface area

8. Order of reaction is found by: (a) Colour (b) Experiment (c) Balancing equation (d) Molar mass

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why do not all collisions lead to reaction?

2. How does temperature affect the Boltzmann distribution?

3. Why does powdered calcium carbonate react faster than chips?

4. How does a catalyst increase reaction rate?

5. How does a rate law show the order of a reaction?

6. How does pressure affect the rate of a gaseous reaction?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Explain rate of reaction, rate constant and collision theory. **(4)**

(b) Explain effects of temperature, concentration, surface area and catalyst using Boltzmann distribution. **(4)**

Q.4 (a) Explain rate law and zero, first and second order reactions. **(4)**

(b) Explain activation energy and effective collisions. **(4)**

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